EEE – 3210

Micro-processor Unit Lab

Assignment

Serial Monitor

1. Take User input Radius (r) as Float data. Write a program to calculate the area of the circle.

2. Take 3 numbers as User Inputs (data type INT). Write a program to find out the minimum among these 3 numbers.

3. Consider an array of 4 elements (initialize inputs in the code). Write a program that calculates the average of these 4 numbers.

4. Take 2 inputs (integer) from user. Then ask user what arithmetic operation they want to do with these number (take as a String Input). Then show the output result.

```
Frist Input: 3
Second Input: 2
Operation: Add
Output = 5

First Input: 16
Second Input: 4
Operation: Division
Output = 4

First Input: 5
Second Input: 7
Operation: Multiplication
Output = 35
```

5. Take Student ID as a user input. From ID, you have to find out the relevant information such as Admission year, Term, Department and Roll No.

```
Student ID: 190205123
Admission Year: 2019
Term: Fall
Department: EEE
Roll. No.: 123

Student ID: 170107089
Admission Year: 2017
Term: Spring
Department: IPE
Roll. No.: 089
```

6. Design a Binary Encoder (One-Hot encoded). It will take 4-digit binary input from user and show the equivalent decimal number. It will also check whether user is giving correct input or not.

```
Input: 0100
Output: 10 (D2)

Input: 0010
Output: 01 (D1)

Input: 1001
Output: Wrong
```